REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 12 Advanced | Module 22 Lesson 1 | A Particle Model of Waves

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: $E_{photon} = hf = hc/\lambda$; $KE_{max} = hf - \varphi$; $f_0 = \varphi/h$; $p_{photon} = h/\lambda$. $h = 6.626 \times 10^{-34}$ J s; $c = 3.00 \times 10^{8}$ m/s; 1 eV = 1.602×10^{-19} J.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 46–58.

P01. A photon is:

A. An electron inside a metal B. A quantum of electromagnetic radiation
C. A classical orbit of an atom D. A neutron with charge

Answer

B. A quantum of electromagnetic radiation.

P02. Below a metal's threshold frequency, increasing light intensity alone causes:

A. Greater maximum electron kinetic energy B. Emission at every possible intensity
C. No photoelectron emission in the ideal single-photon model D. The threshold frequency to become zero

Answer

C. No photoelectron emission in the ideal single-photon model.

P03. Above threshold, increasing frequency with intensity otherwise controlled increases:

A. Electron rest mass B. Metal atomic number
C. Planck's constant D. Maximum photoelectron kinetic energy

Answer

D. Maximum photoelectron kinetic energy.

P04. The slope of a KEmax versus frequency graph is:

A. Planck's constant h B. The work function directly
C. The speed of light D. The electron charge in coulombs only

Answer

A. Planck's constant h.

P05. The Compton effect supports the idea that photons possess:

A. Electron rest mass B. Momentum
C. A fixed electric charge D. Zero energy

Answer

B. Momentum.

P06. Find the energy of a photon of frequency 2×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

A. 2.65×10^{-19} J B. 6.63×10^{-20} J
C. 1.33×10^{-19} J D. -1.33×10^{-19} J

Answer

C. 1.33×10^{-19} J. $E = hf = (6.626 \times 10^{-34})(2 \times 10^{14}) = 1.33 \times 10^{-19}$ J.

P07. Find the energy of a photon of wavelength 500 nm. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

A. 7.95×10^{-19} J B. 1.99×10^{-19} J
C. -3.98×10^{-19} J D. 3.98×10^{-19} J

Answer

D. 3.98×10^{-19} J. $E = hc/\lambda = (6.626 \times 10^{-34})(3.00 \times 10^{8})/(5 \times 10^{-7}) = 3.98 \times 10^{-19}$ J.

P08. A photon has energy 4 eV and the metal work function is 2.0 eV. Find the maximum photoelectron kinetic energy.

A. 2 eV B. 4 eV
C. 1 eV D. -2 eV

Answer

A. 2 eV. $KE_{max} = E_{photon} - \varphi = 4 - 2.0 = 2$ eV.

P09. A metal work function is 3.84×10^{-19} J. Find its threshold frequency. Use $h = 6.626 \times 10^{-34}$ J s.

A. 1.16×10^{15} Hz B. 5.8×10^{14} Hz
C. 2.9×10^{14} Hz D. -5.8×10^{14} Hz

Answer

B. 5.8×10^{14} Hz. $f_0 = \varphi/h = (3.84 \times 10^{-19})/(6.626 \times 10^{-34}) = 5.8 \times 10^{14}$ Hz.

P10. Calculate the momentum of a photon with wavelength 500 nm. Use $h = 6.626 \times 10^{-34}$ J s.

A. 2.65×10^{-27} kg m/s B. 6.63×10^{-28} kg m/s
C. 1.33×10^{-27} kg m/s D. -1.33×10^{-27} kg m/s

Answer

C. 1.33×10^{-27} kg m/s. $p = h/\lambda = (6.626 \times 10^{-34})/(5 \times 10^{-7}) = 1.33 \times 10^{-27}$ kg m/s.

P11. Define the photoelectric effect.

Answer

The emission of electrons from a material when incident electromagnetic radiation supplies sufficient energy per photon.

P12. Define work function.

Answer

The minimum energy needed to remove an electron from a material's surface.

P13. Define threshold frequency.

Answer

The minimum incident-light frequency capable of producing photoemission from the material in the single-photon model.

P14. Explain how light intensity affects photoelectric current above threshold.

Answer

Greater intensity at the same frequency supplies more photons per second, so more electrons can be emitted per second, subject to collection conditions.

P15. What is an electron volt?

Answer

The energy gained by a charge of magnitude e moving through a potential difference of 1 V; 1 eV = 1.602×10^{-19} J.

P16. Find the energy of a photon of frequency 3×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$E = hf = (6.626 \times 10^{-34})(3 \times 10^{14}) = 1.99 \times 10^{-19}$ J.

P17. Find the energy of a photon of frequency 4×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$E = hf = (6.626 \times 10^{-34})(4 \times 10^{14}) = 2.65 \times 10^{-19}$ J.

P18. Find the energy of a photon of wavelength 550 nm. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

Answer

$E = hc/\lambda = (6.626 \times 10^{-34})(3.00 \times 10^{8})/(5.5 \times 10^{-7}) = 3.61 \times 10^{-19}$ J.

P19. Find the energy of a photon of wavelength 600 nm. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

Answer

$E = hc/\lambda = (6.626 \times 10^{-34})(3.00 \times 10^{8})/(6 \times 10^{-7}) = 3.31 \times 10^{-19}$ J.

P20. A photon has energy 4.5 eV and the metal work function is 2.0 eV. Find the maximum photoelectron kinetic energy.

Answer

$KE_{max} = E_{photon} - \varphi = 4.5 - 2.0 = 2.5$ eV.

P21. A photon has energy 5 eV and the metal work function is 2.0 eV. Find the maximum photoelectron kinetic energy.

Answer

$KE_{max} = E_{photon} - \varphi = 5 - 2.0 = 3$ eV.

P22. A metal work function is 4.17×10^{-19} J. Find its threshold frequency. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$f_0 = \varphi/h = (4.17 \times 10^{-19})/(6.626 \times 10^{-34}) = 6.29 \times 10^{14}$ Hz.

P23. A metal work function is 4.49×10^{-19} J. Find its threshold frequency. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$f_0 = \varphi/h = (4.49 \times 10^{-19})/(6.626 \times 10^{-34}) = 6.78 \times 10^{14}$ Hz.

P24. Calculate the momentum of a photon with wavelength 550 nm. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$p = h/\lambda = (6.626 \times 10^{-34})/(5.5 \times 10^{-7}) = 1.2 \times 10^{-27}$ kg m/s.

P25. Calculate the momentum of a photon with wavelength 600 nm. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

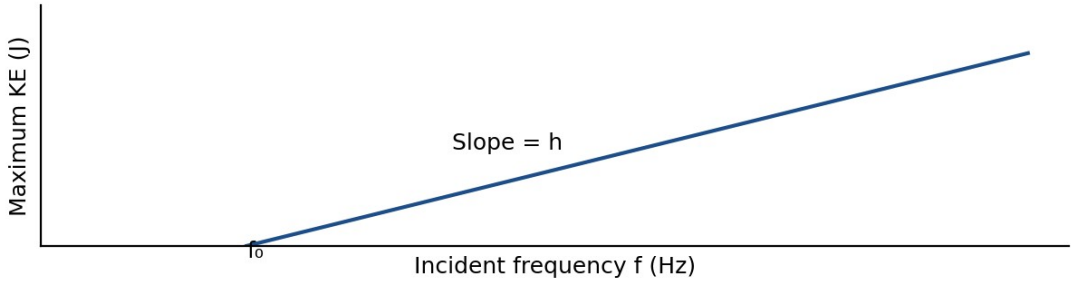
$p = h/\lambda = (6.626 \times 10^{-34})/(6 \times 10^{-7}) = 1.1 \times 10^{-27}$ kg m/s.

P26. A metal has work function 2.0 eV and is illuminated by 3.5 eV photons. Find KEmax and explain the effect of doubling intensity.

Answer

KEmax = 3.5 − 2.0 = 1.5 eV. Doubling intensity ideally increases emitted-electron rate and current but not KEmax at fixed photon frequency.

P27. Sketch a KEmax–frequency graph for a metal. Label threshold frequency and explain slope.



Answer

Draw a straight line in the emitting region with slope h. Its frequency-axis intercept is f0 = φ/h. Below f0 there is no photoemission in the ideal model.

P28. Why did immediate emission above threshold challenge a purely classical energy-accumulation explanation?

Answer

Electrons obtain energy from individual photons. Emission can occur without the long accumulation delay predicted for weak classical illumination, provided hf exceeds the work function.

P29. Describe how a graph can be used to estimate Planck's constant.

Answer

Measure KEmax for several incident frequencies above threshold, plot KEmax in joules against frequency in hertz and find the best-fit slope h.

P30. Explain the wavelength increase in a Compton-scattered photon qualitatively.

Answer

The photon transfers energy and momentum to an electron. A lower-energy scattered photon has lower frequency and hence a longer wavelength.